From Lab to Field: Diagnosing the Internal-to-Field Accuracy Gap in Sugarcane and Pigeon Pea Disease Classification

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***Abstract*—*Models that look excellent on data gathered the same way as their training set often stumble on real field photographs, yet internal validation accuracy is still the figure most people use to judge them. To measure how large that gap is, we ran 52 transfer-learning experiments (23 on Sugarcane, 29 on Pigeon Pea). Every model was trained on the controlled v1 dataset and then tested on v2, an independently collected set that better reflects field conditions. Across eleven architecture families we varied one factor at a time: augmentation, class-weighted loss, dropout with frozen batch-norm statistics, Mixup, and Sharpness-Aware Minimization (SAM). For Sugarcane, internal and field accuracy were negatively associated (Pearson $r = -0.74$, Spearman $\rho = -0.73$, both $p < 0.001$); for Pigeon Pea they were statistically uncorrelated (Pearson $r = 0.14$, $p = 0.51$; Spearman $\rho = 0.05$, $p = 0.82$). The Sugarcane baselines scored 95.55–96.73% internally but only 31.55–38.87% on v2, a gap of 57–65 points that switching architectures did not close. The best Sugarcane field result came from a class-weighted CSPDarkNet53 (66.48% on v2), and the best Pigeon Pea result came from ConvNeXt-Base with SAM (71.31%), which gave up fewer than two points of internal accuracy. In our experiments, rebalancing the classes and using sharpness-aware optimization narrowed the domain gap most reliably, while internal validation accuracy on its own was a poor guide to field performance.*

***Index Terms*—*Crop disease classification, domain adaptation, domain shift, transfer learning, precision agriculture, class imbalance.*

I. INTRODUCTION

Smallholder farmers need a quick and dependable way to identify crop disease, and at present they rarely have one. In Maharashtra, Sugarcane suffers from Red Rot, Rust, Mosaic and Yellow Leaf Disease, while Pigeon Pea is hit by Phytophthora Blight, Fusarium Wilt, Pod Borer damage and Sterility Root Rot. These problems cut yields season after season, and manual inspection cannot keep pace. CNNs pretrained on large natural-image datasets [1], [2] transfer well to agricultural classification. The catch is that most published results are measured on held-out images from the same collection pipeline used for training. When deployment images differ in camera, lighting, background or framing, as field photographs usually do, that kind of test overstates real-world performance. This mismatch is known as domain shift.

This study was carried out under the CHANAKYA Fellowship Programme at TIH IIT Bombay, in support of the MahaVISTAAR AI-based agricultural advisory platform. We train on the v1 dataset, which is government-sourced imagery, and evaluate on v2, an independently collected set closer to unconstrained field photography. The gap between the two results is what we set out to study. Our contributions are threefold. First, we characterize domain shift across two crops using 52 controlled experiments that span several architectures and training interventions. Second, we separate the interventions that actually narrow the domain gap from those that only raise internal accuracy. Third, we show that internal accuracy is a weak predictor of field accuracy, which argues for making cross-domain evaluation a standard step rather than an optional extra.

II. RELATED WORK

ImageNet-pretrained backbones are well established for plant disease classification. Residual networks [1], efficient scaling [2], modernized convolutional designs [3] and self-supervised foundation models [4] all perform strongly on benchmarks, and regularizers such as SAM [5] and Mixup [6] are known to improve generalization in general. Yet most agricultural classification studies still report in-distribution accuracy only, and controlled studies that test several crops against field-deployment data are rare. A 2026 review of 108 AI-based crop disease studies reports that models routinely exceed 95% accuracy under controlled conditions but drop noticeably on field imagery, and it names the lack of cross-domain validation as a persistent gap [7]. Yang et al. [8] address the same lab-to-field problem with cross-domain few-shot learning. They also observe a sharp decline, but they approach it through meta-learning and do not compare interventions one by one. Daphal and Koli [9] and Kunduracioglu and Pacal [10] report high Sugarcane accuracy, but only in-distribution. The nearest Pigeon Pea precedent is Bhagat et al. [11], who built a field-collected dataset covering eighteen Indian states; however, they evaluate within that dataset and do not train on lab images and test on field images. To our knowledge, none of this earlier work isolates the internal-versus-field divergence examined here.

III. DATASET DESCRIPTION

A. Sugarcane Dataset

The Sugarcane training set is the v1 collection (SC_dataset_v1) [12] from the Government of Maharashtra. It holds roughly 10,500 RGB leaf images in 7 classes (Healthy, Mosaic, Red Rot, Redrot Rust, Rust, Unknown and Yellow Leaf Disease), with 488 to 1,988 images per class. For evaluation we use v2 [13], a national collection assembled from state agricultural universities. It comes as a non-validated split of 1,195 images and an expert-validated split of 355 images, and both cover only 4 shared classes. Unless we say otherwise, all v2 numbers in this paper come from the validated split. The difference between 7 training classes and 4 test classes is itself a

structural source of domain shift, in addition to the visual differences between the two pipelines.

B. Pigeon Pea Dataset

The Pigeon Pea data also come from the v1 collection (PP_dataset_v1) and contain RGB leaf and plant images in 7 classes: Healthy, Phytophthora Blight, Pod Borer, Root Rot, Unknown, Webbed Leaves and Wilt. Pod Borer (870 images) and Wilt (570) are the largest classes, and Root Rot (300) is the smallest. Evaluation uses the validated v2 split of 366 images, which covers the same 4 shared classes. On-the-fly augmentation brought the effective training count to 500–1,000 images per class.

IV. METHODOLOGY

A. Preprocessing and Augmentation

All images are resized to 224×224 and normalized with ImageNet channel statistics (mean 0.485, 0.456, 0.406; standard deviation 0.229, 0.224, 0.225). Augmentation is applied on the fly and includes random horizontal flips, rotation, color jitter, Gaussian blur, shear, translation, random erasing and random perspective. For Pigeon Pea Exp 22–29 (Table II) we also apply CLAHE [14] before normalization. CLAHE equalizes contrast within small tiles instead of across the whole image, which suits field photographs because their illumination varies much more than that of curated images. Later experiments additionally remove near-duplicates using perceptual hashing and add text-overlay augmentation designed to resemble v2-style images.

B. Two-Phase Transfer Learning

Training proceeds in two stages. In Stage 1 the backbone stays frozen and only the classification head is trained, with early stopping once validation performance has stopped improving for 5–7 epochs. In Stage 2 we unfreeze the higher-level layers (typically layer3 and layer4 in ResNet-family models) and train them together with the head at a lower learning rate. The lower rate protects the transferable features from catastrophic forgetting, while the unfrozen layers can still adapt to the texture and color of the target crops.

C. Loss, Optimization, and Regularization

The baselines use cross-entropy loss with the Adam optimizer. Later experiments change this in several ways. Class-weighted cross-entropy is added to offset class imbalance, and AdamW [15] with weight decay serves as a regularizer. SAM [5] steers training toward flatter minima, which are associated with better out-of-distribution behavior, and Mixup [6] is tested as an extra regularizer. We also place dropout before the FC head and freeze the batch-normalization statistics during Stage 2. This combination was introduced to recover the internal accuracy lost when class weighting is applied: the frozen statistics preserve the v1-derived normalization while the discriminative weights continue to update.

D. Evaluation Protocol

Internal accuracy is the validation accuracy on the held-out v1 split. Cross-domain accuracy is the overall accuracy on the validated v2 split, and we treat it as the main indicator of readiness for deployment. Both numbers are reported side by side throughout, because comparing them is the basis of the analysis in Section VI.

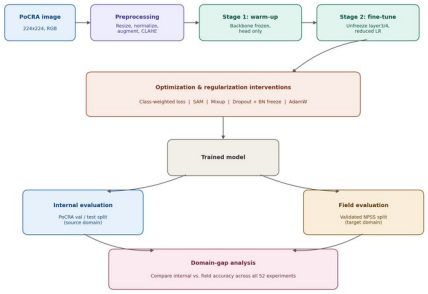


Fig. 1. System overview: preprocessing, two-phase transfer learning, the optimization and regularization interventions under comparison, and the paired internal and field evaluation.

V. RESULTS

A. Sugarcane

Table I gives internal validation, internal test and field-image test accuracy for all 23 Sugarcane experiments, and Fig. 2 plots the three measures for each row. The baselines reach 95.55–96.73% internal accuracy but score only 31.55–38.87% on the field set, so the baseline domain gap is roughly 57–65 points. A MobileNetV2 [16] baseline that was evaluated only on the non-validated split did even worse (23.77%). We leave it out so that every row is judged on the same evaluation data.

TABLE I
SUGARCANE EXPERIMENTS: INTERNAL AND FIELD-IMAGE TEST ACCURACY (VALIDATED FIELD SET, 355 IMAGES)

Exp	Model	Technique	Val. (%)	Test (%)	Field (%)
Base	EfficientNet-B0	Baseline pipeline	96.73	—	35.77
2	ResNet50	Baseline pipeline	95.55	96.37	38.87
4	ResNet34	Baseline pipeline	96.16	96.83	31.55
7	ResNet50	Reduced LR + AdamW + longer patience	96.16	95.77	25.63
9	ResNet50	+ Augmentation	94.47	94.11	41.13
11	ResNet34	+ Augmentation	91.55	94.26	48.17
13	ResNet50	+ Aug. + Class Weights	86.64	86.86	65.07
15	ResNet34	+ Aug. + Class Weights	84.33	86.10	65.63
17	ResNet50	+ Aug. + Dropout + BN Freeze	93.70	94.56	45.07
19	ResNet34	+ Aug. + Dropout + BN Freeze	89.55	90.63	47.61
21	ResNet50	+ Dropout/BN Frz. + Mixup	95.39	95.92	36.34
23	ResNet34	+ Dropout/BN Frz. + Mixup	93.24	95.92	47.04
25	ResNet50	+ Dropout/BN Frz. + SAM	94.62	96.53	43.94
28	EfficientNet-B7	+ Dropout/BN Frz.	89.86	89.12	40.00
30	EfficientNet-B7	+ Dropout/BN Frz. + SAM	94.01	92.75	44.51
32	DenseNet201	+ Dropout/BN Frz. + SAM	96.16	96.83	35.77
34	ConvNeXt-Base	+ Dropout/BN Frz.	95.24	97.13	44.51
36	ConvNeXt-Base	+ Dropout/BN Frz. + SAM	95.55	97.28	47.32
37	CSPDarkNet53	+ Dropout/BN Frz.	91.86	93.81	44.79
38	Swin Transformer	+ Augmentation	94.93	96.37	43.38
39	ConvNeXt-Base	+ Dropout/BN Frz. + Class Weights	94.93	96.83	54.65

Exp	Model	Technique	Val. (%)	Test (%)	Field (%)
40	CSPDarkNet53	+ Dropout/BN Frz. + Class Weights	87.56	88.07	66.48
41	DINOv2	+ Aug. + Class Weights	95.24	96.37	57.46

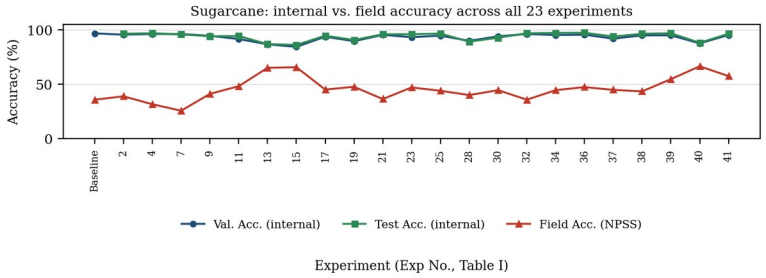


Fig. 2. Validation, test and field (v2) accuracy for each Sugarcane experiment in Table I. Internal accuracy stays high throughout, while field accuracy moves on its own and rises clearly only for the class-weighted runs (Exp 13, 15, 40).

Augmentation by itself (Exp 9, 11) raises v2 accuracy into the low-to-high forties at little cost in internal accuracy. This makes sense, since it shows the model more geometric and photometric variation without changing the label distribution. Adding class weighting on top (Exp 13, 15) gives a much bigger jump, to 65.07% and 65.63%, but internal accuracy falls by 7–10 points. One reading is that unweighted training on v1 produces a decision boundary shaped partly by class frequency instead of by disease-relevant features. Regularization alone tells a different story. Dropout with BN freezing (Exp 17, 19), Mixup (Exp 21, 23) and SAM (Exp 25) all keep internal accuracy high (89–95%), yet none lifts v2 accuracy beyond the high forties. They help in the source domain and do relatively little about the shift to the target domain. Mixup on ResNet50 (Exp 21) even lowered v2 accuracy compared with plain dropout (45.07% to 36.34%), so adding more regularization does not automatically help.

Architecture matters mainly when it is combined with class weighting. Without weighting, the stronger backbones all land in the mid-forties on v2: ConvNeXt-Base 44.51–47.32%, CSPDarkNet53 44.79%, Swin 43.38%, EfficientNet-B7 40.00–44.51% and DenseNet201 35.77%, even though their internal accuracy is 89–96%. Once class weighting is added, CSPDarkNet53 reaches 66.48%, the best cross-domain result in the study and more than 21 points above its unweighted run. It does this with the lowest internal accuracy among the modern backbones (87.56%). DINOv2 comes next at 57.46%, the strongest foundation-model result. It trains only a fine-tuned head on frozen self-supervised features, which suggests that self-supervised pretraining gives representations that are less tied to the collection conditions of any single dataset.

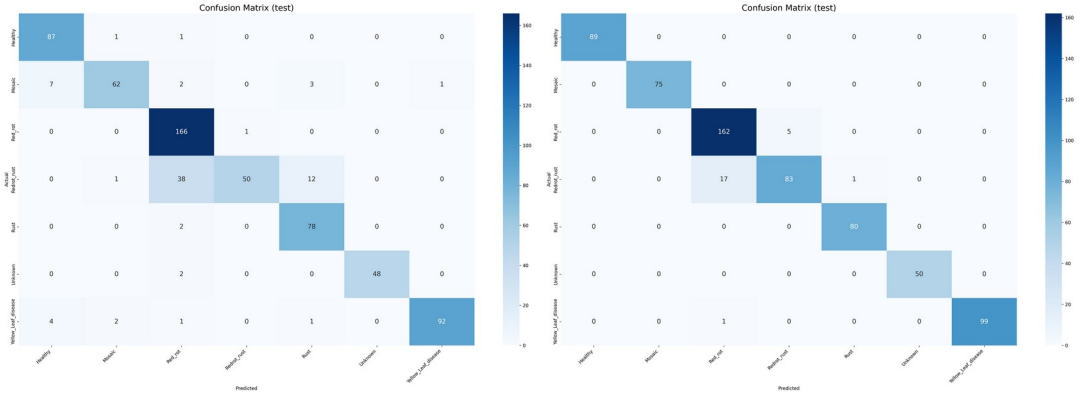


Fig. 3. Confusion matrices on the internal test split for the two highlighted Sugarcane results. Left: CSPDarkNet53 + Dropout/BN Freeze + Class Weights (Exp 40), the best field accuracy at 66.48% (88.07% internal test). Right: DINOv2 + Augmentation + Class Weights (Exp 41), the best foundation-model result at 57.46% field accuracy (96.37% internal test).

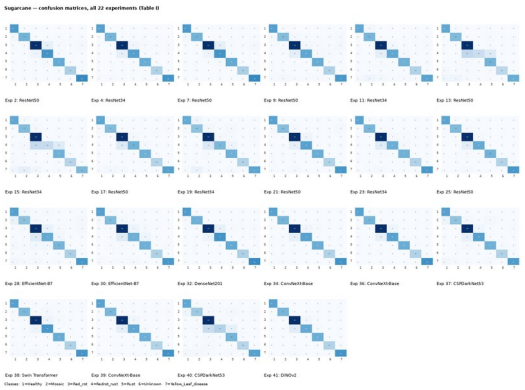


Fig. 4. Confusion matrices (internal test split) for the 22 Sugarcane experiments that have a trained checkpoint available (Table I), shown as a compact reference next to Fig. 3.

B. Pigeon Pea

Table II gives the results for all 29 Pigeon Pea experiments, and Fig. 5 plots the three accuracy measures for each one.

TABLE II
PIGEON PEA EXPERIMENTS: INTERNAL AND FIELD-IMAGE TEST ACCURACY (VALIDATED FIELD SET, 366 IMAGES)

Exp	Model	Technique	Val. (%)	Test (%)	Field (%)
1	ResNet34	Baseline pipeline	90.66	88.95	55.74
2	ResNet50	Two-phase fine-tuning, patience 7	92.03	88.41	65.57
3	VGG16	+ Augmentation	84.02	81.08	57.38
4	VGG19	+ Augmentation	76.03	74.86	25.41
5	DenseNet201	+ Dropout/BN Frz. + SAM	90.63	90.54	62.02
6	CSPDarkNet53	+ Dropout/BN Frz.	76.58	77.03	62.02
7	ConvNeXt-Base	+ Dropout/BN Frz.	91.18	91.89	66.94
8	EfficientNet-B0	+ Augmentation	81.82	82.70	61.48
9	ResNet18	+ Augmentation	81.82	84.86	60.93
10	ShuffleNet	+ Augmentation	82.92	82.16	—
11	LiteMDC	+ Augmentation	82.92	65.01	—
12	ConvNeXt-Base	+ Augmentation + SAM	89.26	91.89	71.31
13	CSPDarkNet53	+ Augmentation + SAM	80.72	78.65	52.46
14	DenseNet201	+ Augmentation + SAM	86.23	86.49	62.02
15	EfficientNet-B0	+ Augmentation + SAM	82.92	83.24	67.21
16	LiteMDC	+ Augmentation + SAM	48.48	49.19	—
17	MobileNetV2	+ Augmentation + SAM	77.69	75.68	63.66
18	ResNet18	+ Augmentation + SAM	75.76	75.95	64.75
19	ShuffleNet	+ Augmentation + SAM	69.97	70.27	50.82
20	VGG16	+ Augmentation + SAM	74.38	74.59	52.19
21	VGG19	+ Augmentation + SAM	79.06	79.19	53.01
22	Swin Transformer	+ Augmentation + CLAHE	98.56	99.01	49.73
23	DINOv2	+ Augmentation + CLAHE	98.42	99.01	57.38
24	ConvNeXt-Base	+ Augmentation + CLAHE	95.11	97.16	48.36
25	DenseNet201	+ Augmentation + CLAHE	90.94	93.04	58.20
26	VGG19	+ Augmentation + CLAHE	89.64	91.48	47.81
27	ResNet50	+ Augmentation + CLAHE	88.63	90.48	57.10
28	EfficientNet-B7	+ Augmentation + CLAHE	87.34	90.06	64.21
29	VGG16	+ Augmentation + CLAHE	87.77	89.77	45.63

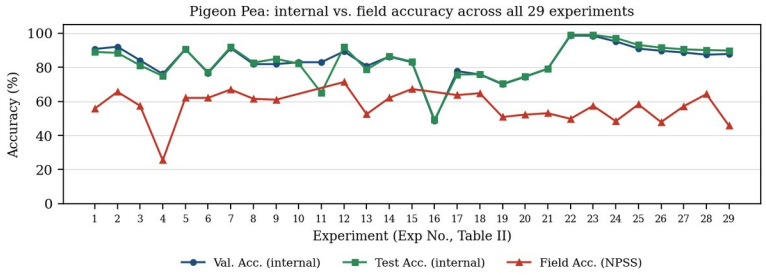


Fig. 5. Validation, test and field (v2) accuracy for each Pigeon Pea experiment in Table II. Field accuracy follows internal accuracy more closely than it does for Sugarcane, yet the best field result (Exp 12) is not the one with the highest internal accuracy.

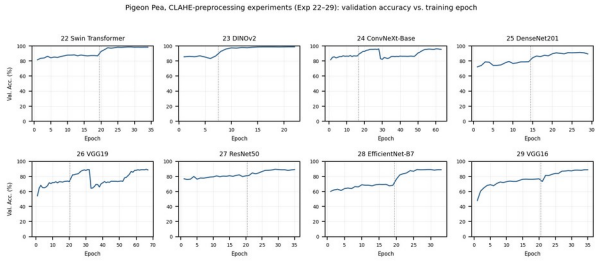


Fig. 6. Validation accuracy against training epoch for the eight CLAHE-preprocessing Pigeon Pea experiments (Exp 22–29). The dotted line marks where Stage 2 fine-tuning begins.

Pigeon Pea shows the same trade-off as Sugarcane, only on a smaller scale. The best cross-domain model is ConvNeXt-Base with SAM (Exp 12), which reaches 71.31% field accuracy with 89.26% internal validation accuracy. It does not have the highest internal accuracy: ResNet50 with two-phase fine-tuning reaches 92.03% internally but only 65.57% in the field, and the CLAHE runs of Swin Transformer [17] and DINOv2 score 98.56% and 98.42% internally while reaching just 49.73% and 57.38% in the field. Adding SAM to ConvNeXt-Base costs under two points of internal accuracy and gains more than four points in the field over the same backbone without SAM (Exp 7). The VGG16 and VGG19 comparison [18] is also telling: the deeper VGG19 has lower internal accuracy and a far lower v2 accuracy (25.41% against 57.38%), which fits the idea that deeper networks overfit more severely to a small, imbalanced training set. DenseNet201 [19] with SAM and CSPDarkNet53 both end at exactly 62.02% despite very different internal accuracies (90.63% and 76.58%). Five of the eight highlighted configurations fall in a narrow 61–

67% band however they were reached, pointing to a generalization ceiling set by the dataset itself (about 3,400 images, one class at only 300) rather than by any particular architecture.

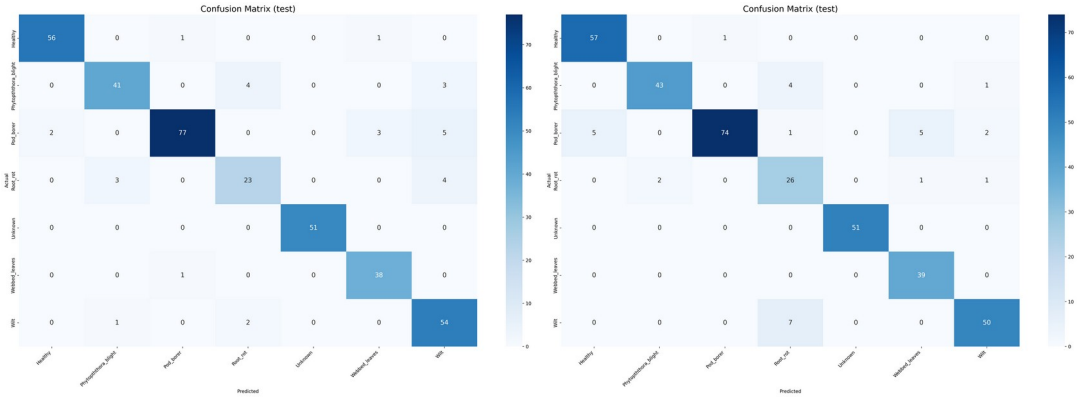


Fig. 7. Confusion matrices on the internal test split for the two highlighted Pigeon Pea results. Left: ConvNeXt-Base + Augmentation + SAM (Exp 12), the highest field accuracy at 71.31% (91.89% internal test). Right: ConvNeXt-Base + Dropout/BN Freeze (Exp 7), 66.94% field accuracy (91.89% internal test).

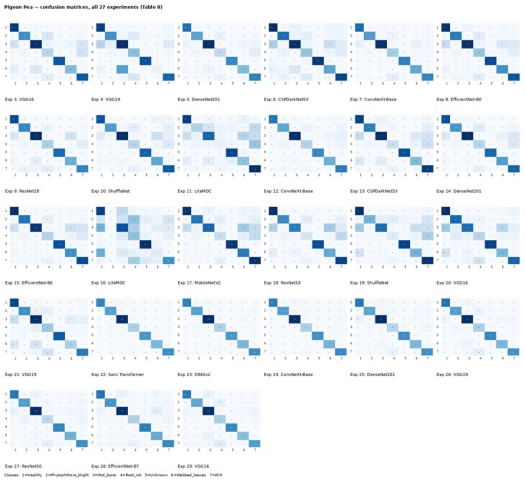


Fig. 8. Confusion matrices (internal test split) for the 27 Pigeon Pea experiments that have a trained checkpoint available (Table II), shown as a compact reference next to Fig. 7.

C. Correlation Between Internal and Field Accuracy

To check whether internal accuracy predicts field accuracy, we computed Pearson and Spearman correlations between v1 validation accuracy and v2 field accuracy for each crop, using every experiment with a reported field accuracy (23 of 23 for Sugarcane; 26 of 29 for Pigeon Pea, excluding Exp 10, 11 and 16, which were not evaluated on v2). Fig. 9 plots both variables. For Sugarcane the correlation is negative and significant (Pearson $r = -0.74$, $p < 0.001$; Spearman $\rho = -0.73$, $p < 0.001$), so higher internal accuracy tends to go with lower field accuracy. This fits the pattern in which regularization and augmentation raise internal accuracy, while class weighting, which usually costs internal accuracy, is what raises field accuracy. For Pigeon Pea the correlation is small and not significant (Pearson $r = 0.14$, $p = 0.51$; Spearman $\rho = 0.05$, $p = 0.82$), so the two measures are effectively decoupled, consistent with the narrower range of interventions and the absence of a class-weighted group. In neither crop is internal accuracy a trustworthy proxy for field accuracy, though the mechanism differs: for Sugarcane they move in opposite directions, and for Pigeon Pea one says nothing reliable about the other.

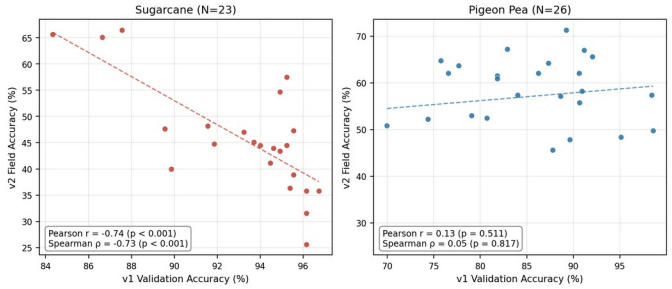


Fig. 9. v1 validation accuracy plotted against v2 field accuracy for every experiment with a reported field result: Table I (Sugarcane, N=23) and Table II (Pigeon Pea, N=26). Pearson r and Spearman ρ are shown in each panel.

VI. DISCUSSION

Both crops reach a comparable cross-domain ceiling, but through different levers (Sections V-A and V-B). We attribute this to dataset size and class distribution. Sugarcane's larger set can take aggressive class rebalancing without its internal accuracy collapsing, yet narrowing the domain gap still costs a large internal sacrifice. Pigeon Pea's smaller, imbalanced set responds better to flatter-minima optimization (SAM), because a small change improves generalization at little internal cost. Standard transfer learning assumes that source and target share the same feature geometry. The v1 and v2 data violate that assumption in three ways: acquisition conditions, the 7-versus-4 class-label mismatch, and class balance. The baseline models address none of these, which explains why they show the largest domain gap here.

The architectural results also have a plausible explanation. CSPDarkNet53's cross-stage partial connections [20] were designed to preserve gradient diversity for object detection, which needs spatially localized, background-invariant features. That likely helps here: the diagnostic signal (lesion shape, color and texture) is spatially localized, while backgrounds differ considerably between v1 and v2. This agrees with CSPDarkNet53 gaining more from class weighting than any other backbone we tested. ConvNeXt-Base [3], with its modernized design, larger receptive fields and transformer-inspired components, has enough capacity to fit the internal distribution, and combined with SAM it stays robust enough to carry that capacity into the field domain, making it the strongest backbone for both crops when paired with the right intervention. The correlation analysis (Section V-C) shows the divergence is systematic rather than anecdotal: selecting models on internal accuracy alone, still common in applied agricultural vision, risks choosing models overfit to the quirks of the training pipeline instead of the disease itself.

VII. LIMITATIONS

Several limitations apply. Even our best cross-domain results (66.48% and 71.31%) remain well below internal accuracy, so these models are not ready for unsupervised deployment. The 7-class v1 versus 4-class v2 mismatch limits direct comparison. Both training sets are imbalanced (Sugarcane 488–1,988 images per class; Pigeon Pea 300–870), which affects minority-class performance in ways overall accuracy does not show. The validated v2 splits are small (355 and 366 images), so some differences between rows amount to only a handful of images. Evaluation uses a single external dataset, each configuration was run once with no confidence intervals, and inference latency, model size and on-device memory were outside our scope.

VIII. FUTURE WORK

A natural next step is explicit domain adaptation, such as correlation alignment or adversarial domain-invariant learning, which would target the v1-to-v2 shift directly instead of relying on class weighting and augmentation as indirect proxies. Class weighting was the strongest Sugarcane lever, yet the modern backbones were only tested without it, so combining the two is an obvious experiment. Self-supervised or semi-supervised learning on unlabeled field imagery, with active learning to choose which images to label, could reduce labeling needs. Further external datasets, larger validated splits and repeated trials with confidence intervals would clarify generalization and settle close comparisons such as the identical 62.02% reached by two Pigeon Pea models. Deployment-side work should also examine latency and memory footprint.

IX. CONCLUSION

We presented a 52-experiment study of domain shift in agricultural disease classification, covering two crops and eleven architecture families, with cross-domain evaluation as the main basis for comparison. Internal and field accuracy diverged substantially for both crops, in different statistical patterns (Section V-C): negatively correlated for Sugarcane and uncorrelated for Pigeon Pea. In both cases internal validation accuracy alone proved an unreliable indicator of deployment readiness. Class-weighted training was the most effective cross-domain lever for Sugarcane and clearly beat architecture selection alone, while SAM played a similar role for Pigeon Pea at a much smaller cost in internal accuracy. Regularization-only methods such as Mixup preserved internal accuracy but did not meaningfully close the field gap. We therefore recommend that cross-domain evaluation on independently collected field data become standard practice in applied agricultural computer vision.

ACKNOWLEDGMENT

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